

Assignment - 1

NAME - ABHINANDAN JAIN

ROLL NO. - 25EARIT002

Question 1

What is a variable in C/C++?

Ans: variable is a named storage location in the computer's memory used to hold data that a program can manipulate

Question 2

- Why do we use variables in programs?
- Ans: variables are used as named containers to temporarily store, manipulate, and retrieve data within a computer's memory during a program's execution

Question 3

- What is the difference between a variable name and its value?
- Ans: The variable name is the identifier, while the value is the actual data stored in it.

Question 4

- Which of the following is a valid variable name? a) 2num b) total_marks c) for d) my marks
- Ans: b) total_marks

Question 5

- Why is Age different from age?
- Ans: C++ is case-sensitive, so Age and age are different variables

Question 6

- Can a variable name contain spaces? Why?
- Ans: No. Variable names cannot contain spaces.

Question 7

- Can a variable name start with an underscore (`_`) ?
- Ans: Yes , variable names can start with an underscore . example: `_count`

Question 8

- What happens if two variables have the same name in the same scope?
- Ans: The compiler generates an error because variable names must be unique within the same
- scope

Question 9

- Write three meaningful variable names for storing student information.
- ANS : studentName, studentAge, studentMarks

Question 10

- Which naming convention is better and why: studentMarks or sm?
- Ans: studentMarks because it is descriptive and easy to understand

Question 11

- What is a data type?
- Ans A data type specifies the kind of data a variable can store.

Question 12

- Which data type is used to store whole numbers?
- Ans : int

Question 13

- Which data type is used to store decimal values?
- Ans: float or double

Question 14

- What is the difference between float and double?
- Ans: double provides greater precision than float.

Question 15

- Which data type stores a single character?
- Ans: char

Question 16

- Which data type stores True/False values?
- Ans: bool

Question 17

- What data type is suitable for storing a person's name?
- Ans: string

Question 18

- Predict the output: `int age=18; cout<<age;`
- Ans: 18

Question 19

- Identify the data type: 25, 3.14, 'A', true.
- Ans: 25=int, 3.14=double, 'A'=char, true=bool

Question 20

- What happens if you store a decimal number in an int variable?
- Ans: The decimal part is discarded. Example:
3.14 becomes 3

Question 21

- What is the purpose of cin?
- Ans: cin is used to take input from the user

Question 22

- What is the purpose of cout?
- Ans: cout is used to display output on the screen.

Question 23

- Which symbol is used with cin?
- Ans: >>

Question 24

- Which symbol is used with cout?
- Ans: <<

Question 25

- Explain: `cin >> age;`
- Ans: It takes input from the user and stores it in `age`.

Question 26

- Explain: `cout << age;`
- Ans: It displays the value of age

Question 27

- What is the purpose of endl?
- Ans: endl moves the cursor to the next line.

Question 28

- Write a statement to input a student's marks.
- Ans: `cin >> marks;`

Question 29

- Write a statement to display 'Welcome to C++'.
- Ans: `cout << "Welcome to C++";`

Question 30

- Predict the output: `cout << 'Programming';`
- Ans: Error because single quotes are for a single character.

Question 31

- What is an operator?
- Ans: An operator is a symbol that performs an operation on operands

Question 32

- Which operator is used for addition?
- Ans:+

Question 33

- What is the result of $10 \% 3$?
- Ans: 1

Explanation: The $\%$ (modulus)operator gives the remainder after division.

$$10 \% 3 = 1$$

Because: $10/3=3$ remainder 1.

Question 34

- What is the result of $8 \% 2$?
- Ans:0
- Because:8 is completely divisible by 2,so remainder is 0.

Question 35

- Which operator checks equality?
- Ans:==

Question 36

- What is the difference between = and == ?
- Ans:= assigns a value, while == compares two values.

Question 37

- What is the result of $5 > 3$?
- Ans:True

Question 38

- What does && represent?
- Ans: Logical AND operator. It returns true only when both conditions are true.

Question 39

- What does `||` represent?
- Ans: Logical OR operator. It returns true if at least one condition is true .

Question 40

- What does ! represent?
- Ans: Logical NOT operator.It reverses the result of a condition.

Question 41

- Write a program to print "Hello world"?

- **Problem Statement:**

- Write a program to display "Hello world " on screen.

- **Algorithm:** 1. start the program.
2. Display "Hello world" using cout
3. End the program.

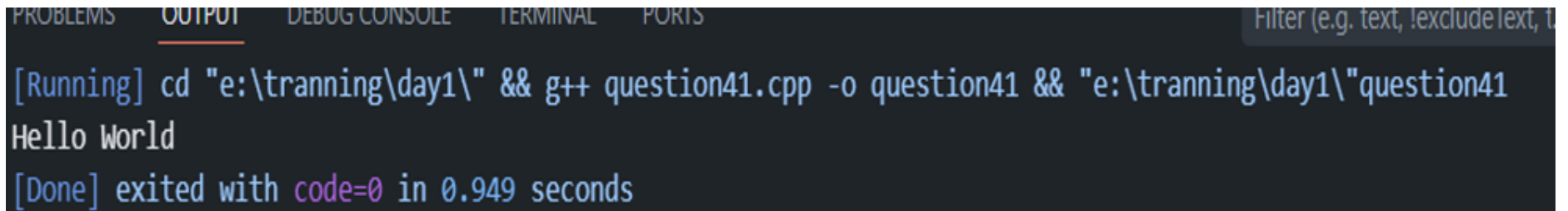
- **C++ Program**

```
#include<iostream>
using namespace std;
int main()
{
    cout<<"Hello world";
    return 0;
}
```

- **Explanation :**

- #include <iostream>: used for input/output operations
- using namespace std; → Allows use of cout without std::

- `cout` → Displays output on screen
- `return 0;` → Ends the program successfully
- **Output**



The screenshot shows a dark-themed interface with tabs at the top: PROBLEMS, OUTPUT (highlighted with an orange underline), DEBUG CONSOLE, TERMINAL, and PORTS. A search filter box on the right contains the text "Filter (e.g. text, !exclude text, t...". The main area displays the following text:

```
[Running] cd "e:\tranning\day1\" && g++ question41.cpp -o question41 && "e:\tranning\day1\"question41
Hello World
[Done] exited with code=0 in 0.949 seconds
```

Question 42

- Write a program to input two numbers and display their sum.
- **Problem Statement**
- Write a program to take two numbers from the user and display their sum.
- **Algorithm**
 1. Start the program
 2. Declare two variables
 3. Take input from user
 4. Add both numbers
 5. Display the result
 6. End the program
- **C++ Program**

```
#include<iostream>
Using namespace std;
int main()
{
int num1,num2,sum;
```

```
cout<<"enter first number:";  
cin>>num1;  
sum=num1 +num2;  
cout<<"sum="<<sum;  
Return 0;  
}
```

Explanation

```
day1> g++ question42.cpp -o question42
```

```
/lib/gcc/mingw32/6.3.0/../../../../mingw32/bin/ld.exe: cannot open output file question42.exe: P  
rror: ld returned 1 exit status
```

```
ber: 10
```

```
mber: 20
```

Question 43

- Write a program to calculate the area of a rectangle.

- **Problem Statement**

Write a program to calculate the area of a rectangle.

- **Formula:**

Area = Length $\times$ Breadth

- **Algorithm**

1. Start the program
 2. Declare variables for length and breadth
 3. Take input from user
 4. Calculate area
 5. Display area
1. End the program

- **C Program**

```
#include<stdio.h>
int main()
{
```

```
float length, breadth, area;  
    printf("Enter length: ");  
    scanf("%f", &length);  
    printf("Enter breadth: ");  
    scanf("%f", &breadth);  
    area = length * breadth;  
    printf("Area of Rectangle = %.2f", area);  
    return 0;  
}
```

Explanation

- scanf() takes input
- length * breadth calculates area
- printf() displays output
- %.2f shows decimal values up to 2 places

```
• >> ./question43  
Enter length: 2  
Enter breadth: 6
```

Question 44

- What formula is used to calculate Simple Interest?
- **Ans:**The formula used to calculate Simple Interest (SI) is:

$$SI = P * R * T / 100$$

Where:

- P = Principal Amount
- R = Rate of Interest
- T = Time
- SI = Simple Interest

Write a program to calculate Simple Interest.

- **Problem Statement**

Write a program to calculate Simple Interest using principal, rate, and time.

- **Algorithm**

1. Start the program
 2. Declare variables
 3. Take principal, rate, and time as input
 4. Apply formula
 5. Display result
1. End the program

C Program

```
#include<stdio.h>
int main()
{
Float principal,rate,time,Sl;
printf("enter principal amont:");
scanf("%f",&principal);
```



```
printf("Enter Rate of Interest: ");
scanf("%f", &rate);
printf("Enter Time: ");
scanf("%f", &time);
SI = (principal * rate * time) / 100;
printf("Simple Interest = %.2f", SI);
return 0;
}
```

Explanation

- User enters principal amount, rate, and time
- Formula is applied to calculate simple interest
- Result is displayed using printf()

```
PS E:\training\day1> gcc question45.c -o question45
• >> ./question45
Enter Principal Amount: 10000
Enter Rate of Interest: 7
Enter Time: 3
Simple Interest = 2100.00
```

Question 46

- Why is a temporary variable used while swapping two numbers?
- **Ans :** A temporary variable is used to safely store one value while swapping two numbers so that the original value is not lost.

QUESTION 47

Swap the values: a=10 b=20

- **Problem Statement**

Write a program to swap two numbers using a temporary variable.

- **Algorithm**

1. Start the program
 2. Declare variables a, b, and temp
 3. Store value of a in temp
 4. Assign value of b to a
 5. Assign value of temp to b
 6. Display swapped values
1. End the program

C++ Program

```
#include<iostream.h>
```

```
Using namespace std;
```

```
int main()
```

```
{
```

```
int a=10,b=20,temp;
```

```
cout << "Before Swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    temp = a;
    a = b;
    b = temp;
    cout << "After Swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    return 0;
}
```

Explanation

- temp = a stores value of a
- a = b copies value of b into a
- b = temp restores original value of a into b
- This swaps both numbers successfully

```
>> ./question4/
Before Swapping:
a = 10
b = 20
After Swapping:
a = 20
```

Question 48

- What formula is used to convert Celsius into Fahrenheit?
- **Ans:** $F = \frac{9}{5}C + 32$
- **Explanation:**
C= temperature in Celsius
F= temperature in fahrenheit

Question 49

- What is the Fahrenheit value of 0°C ?
- **Ans:** using the formula $F = \frac{9}{5}C + 32$
Where $C = 0^{\circ}\text{C}$
- **Calculation:**
$$F = \left(\frac{9}{5} \times 0\right) + 32$$
$$F = 32^{\circ}\text{F}$$
- **Final Answer:**
The Fahrenheit value of 0°C is 32°F .

Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

Answer

Integer division can give incorrect results because integers remove the decimal part.

For example:

5 / 9

This becomes:

0

instead of:

0.555...

So, if integer division is used in the Celsius-to-Fahrenheit formula, the answer may become wrong.

Correct Way

Use float or double for accurate decimal calculations.